Sleep Apnea AHI Estimator (Offline)

Overview. This software provides a fully offline pipeline to detect sleep apnea from ECG recordings derived from the PhysioNet Apnea-ECG dataset. The system converts raw WFDB signals into 60-second epochs, extracts time- and frequency-domain features, applies a trained LightGBM classifier, and aggregates per-epoch predictions to estimate the Apnea–Hypopnea Index (AHI) and severity category. The tool also generates publication-quality evaluation plots and offers a simple web-based interface for demonstration and inspection of results.

Pipeline. 1) Data ingestion from WFDB (.dat/.hea) with minute-wise annotations (.apn).
2) Segmentation into 60-second epochs aligned to annotation timeline.
3) Feature extraction: time-domain statistics (mean, std, RMS, etc.) and frequency-domain descriptors (band powers, spectral centroid).
4) Classification using LightGBM with class balancing.
5) Post-processing to compute AHI and severity (Normal, Mild, Moderate, Severe).
6) Automated report artifacts: ROC curve, Precision–Recall curve, confusion matrix, probability histogram, and timelines.

Performance (Validation). The current model substantially exceeds the provided baseline. Representative results include ROC-AUC ≈ 0.984, PR-AUC ≈ 0.979, F1 ≈ 0.93 at an optimized threshold, accuracy ≈ 0.94, sensitivity ≈ 0.95, and specificity ≈ 0.93. The low Brier score (~0.05) indicates well-calibrated probability estimates. These metrics demonstrate strong separability, high recall for apnea events, and a favorable trade-off between false positives and false negatives.

Usage. The system supports a one-command offline demo (*python src/run_demo.py --offline*) that runs inference and saves all outputs to the *outputs/* directory. A Streamlit-based UI allows interactive inspection of AHI, severity, plots, and prediction timelines using either the default dataset or an uploaded CSV of features.

Dataset and Model. Data are sourced from the PhysioNet Apnea-ECG Database (ODC-By). Labels are derived from minute-wise annotations (A = apnea, N = normal). The classifier is a Gradient Boosted Decision Tree model (LightGBM), chosen for its strong performance, efficiency, and compatibility with feature-based explainability (SHAP).

Disclaimer. This software is intended for research and demonstration only and is not a medical device. It must not be used for clinical diagnosis, treatment, or medical decision-making. Always consult qualified healthcare professionals for medical advice.